Power



File: power.kicad_sch

Vehicle (OBD, CAN, K-Line)



File: vehicle.kicad_sch

Backup battery



File: battery.kicad_sch

MCU (ESP32-C5, USB)



File: mcu.kicad_sch

LTE + GNSS (EC200U)



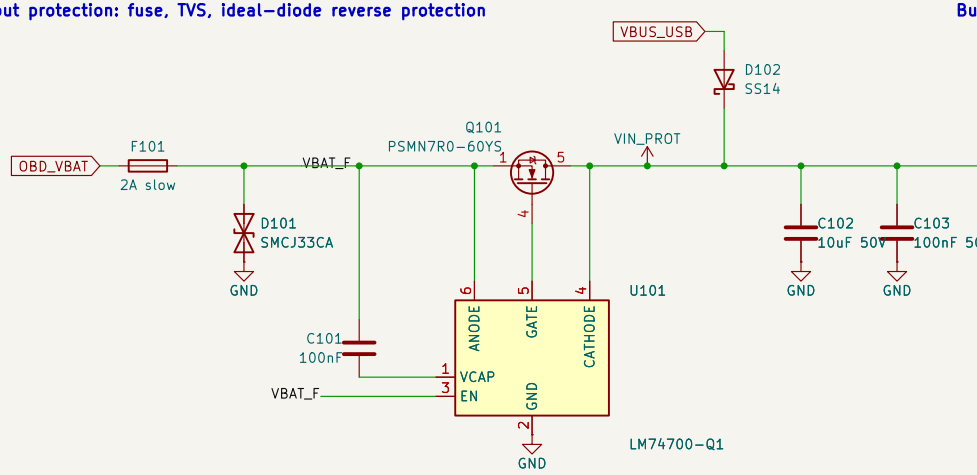
File: lte.kicad_sch

Sheet: /
File: obd-v2x-dongle.kicad_sch

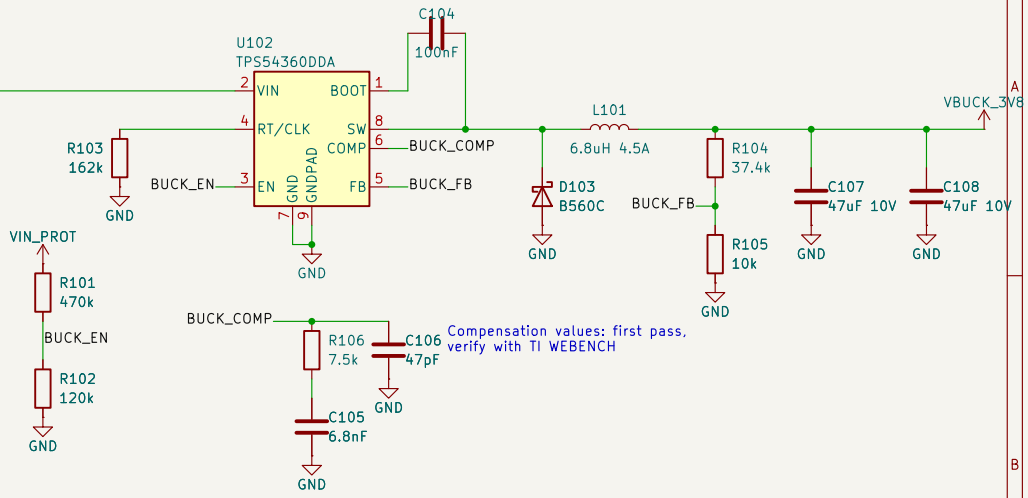
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 1/6

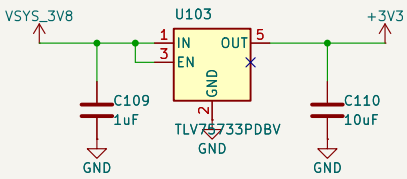
Input protection: fuse, TVS, ideal-diode reverse protection



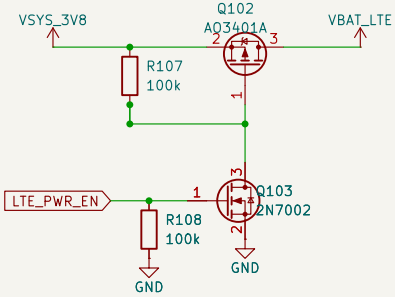
Buck 3.8 V / 3.5 A (TPS54360, 600 kHz)



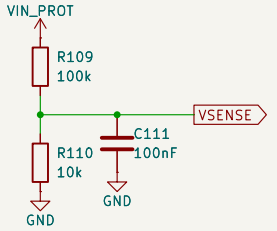
LDO 3.3 V (logic)



LTE modem supply switch (LTE_PWR_EN high = on)



Battery sense (VSENSE = VIN / 11, full scale ~35 V)



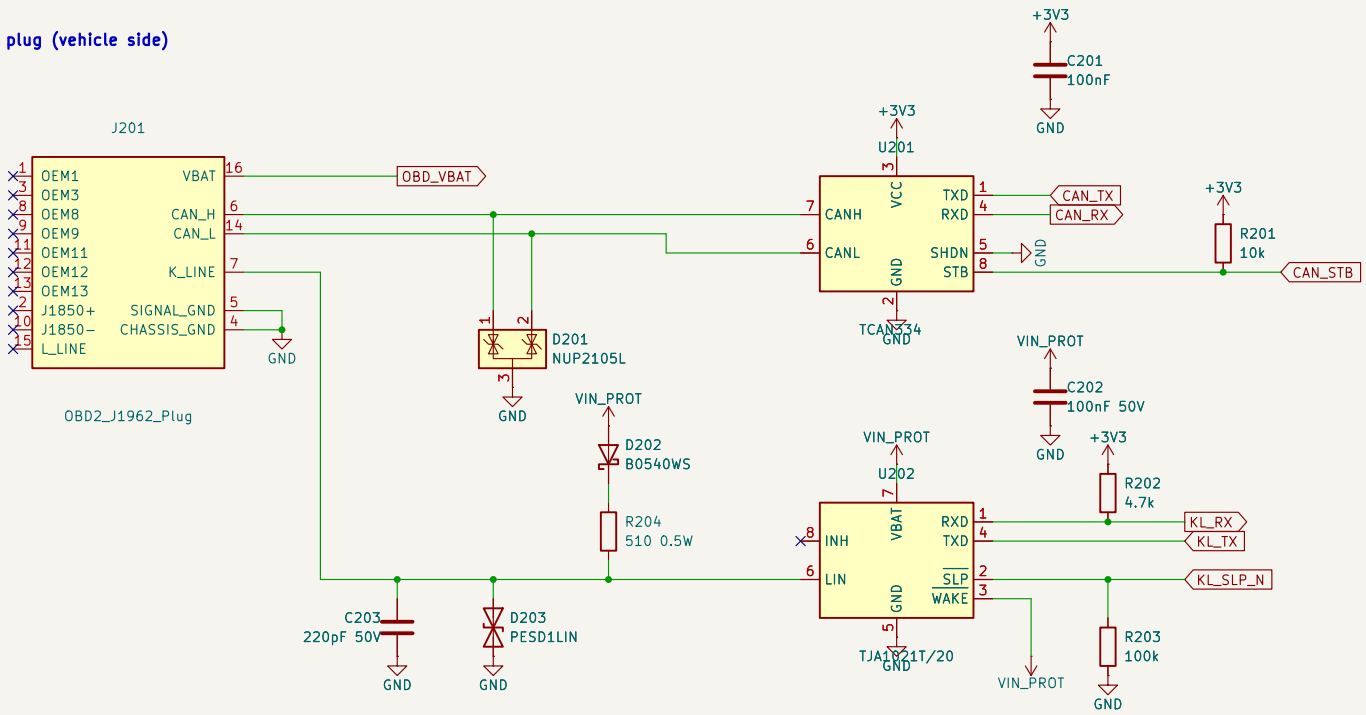
ERC power flags



Sheet: /Power/	
File: power.kicad_sch	
Title: Power: protection, buck 3.8 V, LDO 3.3 V	
Size: A4	Date:
KiCad E.D.A. 10.0.8	**Rev: 0.1**
	Id: 2/6

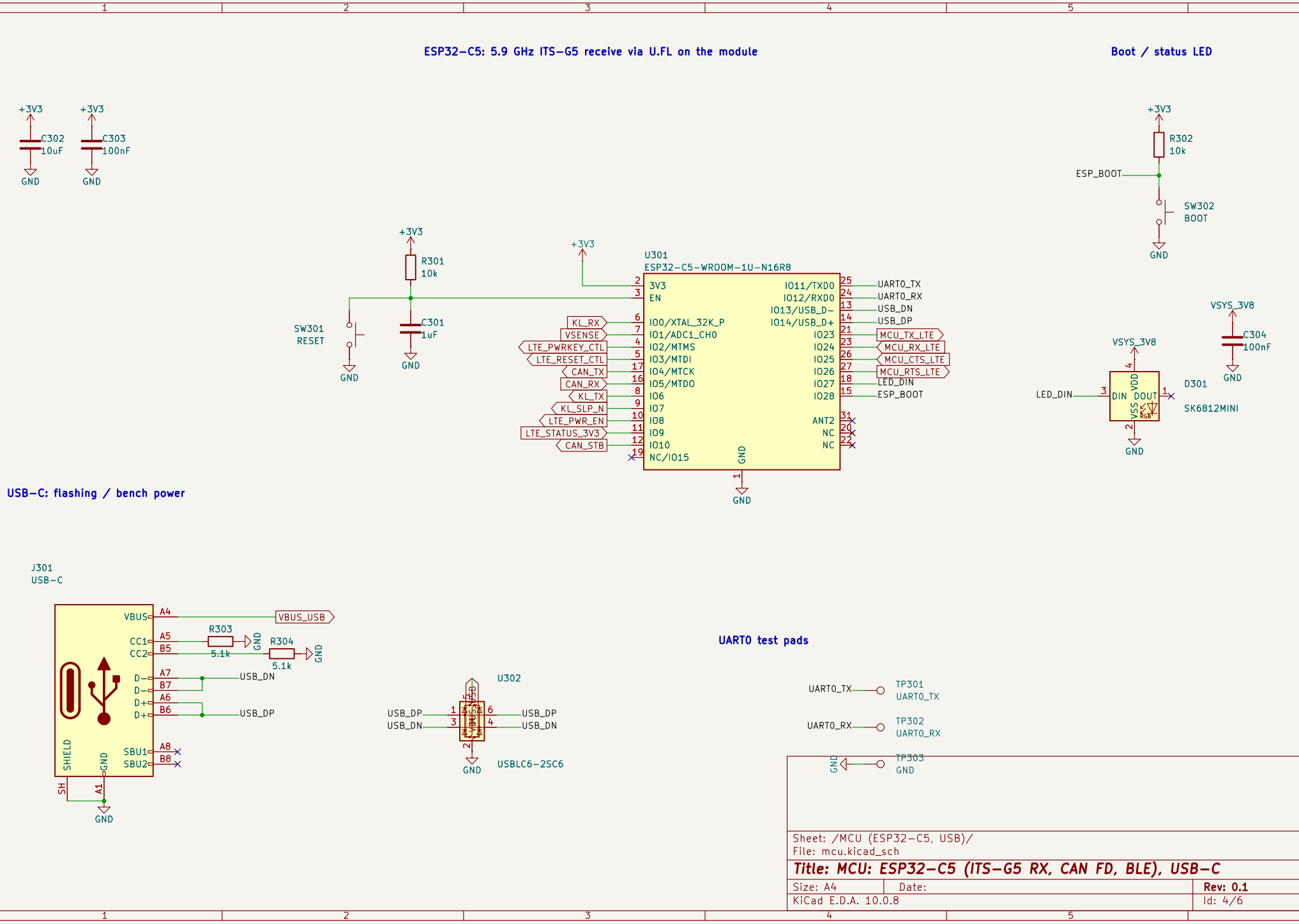
High-speed CAN (ISO 15765), no termination in the dongle

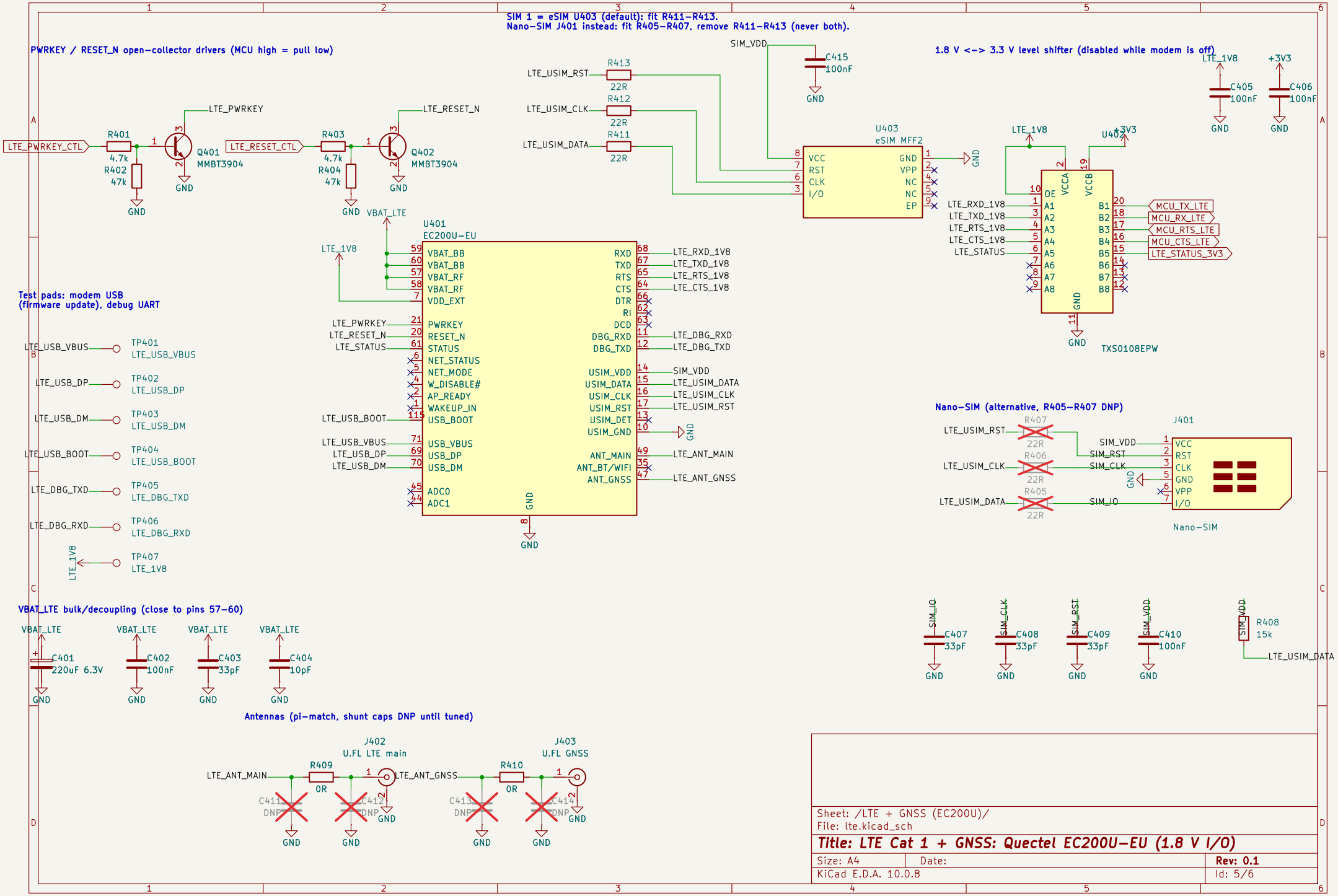
OBD-II plug (vehicle side)



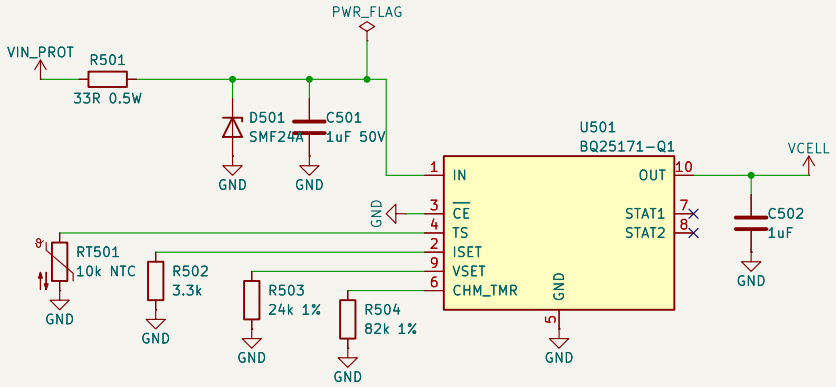
K-Line (ISO 9141 / KWP2000) via LIN transceiver + 510R tester pull-up

Sheet: /Vehicle (OBD, CAN, K-Line)/	
File: vehicle.kicad_sch	
Title: Vehicle interface: OBD-II, CAN, K-Line	
Size: A4	Date:
KiCad E.D.A. 10.0.8	Rev: 0.1
	Id: 3/6


ESP32-C5: 5.9 GHz ITS-G5 receive via U.FL on the module
Boot / status LED
USB-C: flashing / bench power
UART0 test pads
+3V3
C302
10uF
GND
C303
100nF
R301
10k
SW301
RESET
C301
1uF
U301
ESP32-C5-WROOM-1U-N16R8
3V3
EN
IO0/XTAL_32K_P
IO1/ADC1_CH0
IO2/MTMS
IO3/MTDI
IO4/MTCK
IO5/MTDO
IO6
IO7
IO8
IO9
IO10
NC/IO15
IO11/TXD0
IO12/RXD0
IO13/USB_D-
IO14/USB_D+
IO23
IO24
IO25
IO26
IO27
IO28
ANT2
NC
NC
KL_RX
VSENSE
LTE_PWRKEY_CTL
LTE_RESET_CTL
CAN_TX
CAN_RX
KL_TX
KL_SLP_N
LTE_PWR_EN
LTE_STATUS_3V3
CAN_STB
UART0_TX
UART0_RX
USB_DN
USB_DP
MCU_TX_LTE
MCU_RX_LTE
MCU_CTS_LTE
MCU_RTS_LTE
LED_DIN
ESP_BOOT
R302
10k
ESP_BOOT
SW302
BOOT
VSYS_3V8
C304
100nF
D301
SK6812MINI
J301
USB-C
VBUS
CC1
CC2
D-
D+
SBU1
SBU2
SHIELD
SH
R303
5.1k
R304
5.1k
VBUS_USB
U302
USBLC6-2SC6
TP301
UART0_TX
TP302
UART0_RX
TP303
GND
Sheet: /MCU (ESP32-C5, USB)/
File: mcu.kicad_sch
Title: MCU: ESP32-C5 (ITS-G5 RX, CAN FD, BLE), USB-C
Size: A4
Date:
Rev: 0.1
KiCad E.D.A. 10.0.8
Id: 4/6

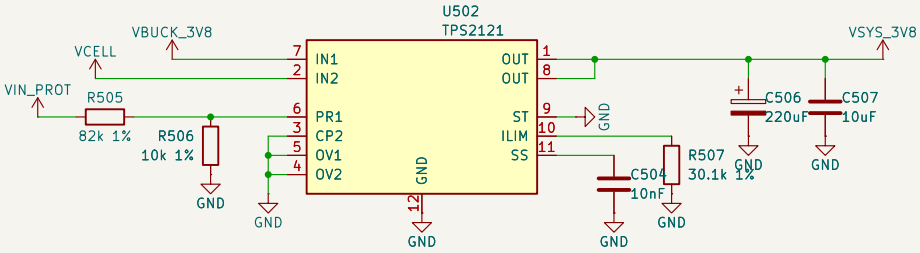

SIM 1 = eSIM U403 (default): fit R411–R413.
Nano-SIM J401 instead: fit R405–R407, remove R411–R413 (never both).
PWRKEY / RESET_N open-collector drivers (MCU high = pull low)
1.8 V <-> 3.3 V level shifter (disabled while modem is off)
Test pads: modem USB
(firmware update), debug UART
Nano-SIM (alternative, R405–R407 DNP)
VBAT_LTE bulk/decoupling (close to pins 57–60)
Antennas (pi-match, shunt caps DNP until tuned)
LTE_PWRKEY_CTL
LTE_RESET_CTL
LTE_PWRKEY
LTE_RESET_N
R401
4.7k
R402
47k
R403
4.7k
R404
47k
Q401
MMBT3904
Q402
MMBT3904
GND
LTE_USIM_RST
LTE_USIM_CLK
LTE_USIM_DATA
R413
22R
R412
22R
R411
22R
SIM_VDD
C415
100nF
U403
eSIM MFF2
VCC
RST
CLK
I/O
GND
VPP
NC
NC
EP
LTE_1V8
+3V3
C405
100nF
C406
100nF
U402
TXS0108EPW
OE
VCCA
VCCB
A1
A2
A3
A4
A5
A6
A7
A8
B1
B2
B3
B4
B5
B6
B7
B8
LTE_RXD_1V8
LTE_TXD_1V8
LTE_RTS_1V8
LTE_CTS_1V8
LTE_STATUS
MCU_TX_LTE
MCU_RX_LTE
MCU_RTS_LTE
MCU_CTS_LTE
LTE_STATUS_3V3
U401
EC200U-EU
VBAT_BB
VBAT_BB
VBAT_RF
VBAT_RF
VDD_EXT
PWRKEY
RESET_N
STATUS
NET_STATUS
NET_MODE
W_DISABLE#
AP_READY
WAKEUP_IN
USB_BOOT
USB_VBUS
USB_DP
USB_DM
ADC0
ADC1
RXD
TXD
RTS
CTS
DTR
RI
DCD
DBG_RXD
DBG_TXD
USIM_VDD
USIM_DATA
USIM_CLK
USIM_RST
USIM_DET
USIM_GND
ANT_MAIN
ANT_BT/WIFI
ANT_GNSS
LTE_DBG_RXD
LTE_DBG_TXD
LTE_ANT_MAIN
LTE_ANT_GNSS
LTE_USB_VBUS
LTE_USB_DP
LTE_USB_DM
LTE_USB_BOOT
TP401
LTE_USB_VBUS
TP402
LTE_USB_DP
TP403
LTE_USB_DM
TP404
LTE_USB_BOOT
TP405
LTE_DBG_TXD
TP406
LTE_DBG_RXD
TP407
LTE_1V8
R407
22R
R406
22R
R405
22R
J401
Nano-SIM
SIM_VDD
SIM_RST
SIM_CLK
SIM_IO
VCC
RST
CLK
GND
VPP
I/O
C401
220uF 6.3V
C402
100nF
C403
33pF
C404
10pF
C407
33pF
C408
33pF
C409
33pF
C410
100nF
R408
15k
J402
U.FL LTE main
J403
U.FL GNSS
R409
0R
R410
0R
C411
DNP
C412
DNP
C413
DNP
C414
DNP
Sheet: /LTE + GNSS (EC200U)/
File: lte.kicad_sch
Title: LTE Cat 1 + GNSS: Quectel EC200U-EU (1.8 V I/O)
Size: A4
Date:
Rev: 0.1
KiCad E.D.A. 10.0.8
Id: 5/6

Li-ion charger, fed from the car side (VIN_PROT)



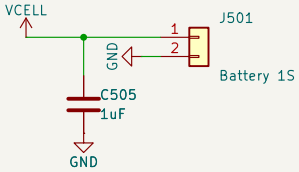
ISET 3.3k: 91 mA charge (kept low: linear from 14 V)
VSET 24k: 4.10 V float CHM_TMR 82k: Li-ion, 10 h timer
TS: 10k NTC next to the cell -> charging only 0..45 C

Priority power mux: buck (IN1) while VIN_PROT > ~9.7 V, else cell (IN2)



ILIM 30.1k: ~3.5 A SS 10 nF
Firmware on battery: LTE only (lower burst current), shut the modem down below ~3.6 V cell

1S Li-ion/LiPo cell, 250-400 mAh, with protection PCM



Sheet: /Backup battery/
File: battery.kicad_sch

Title: Backup battery: 1S Li-ion charger (BQ25171-Q1) + priority power mu

Size: A4	Date:	Rev: 0.1
KiCad E.D.A. 10.0.8		Id: 6/6